



Xarxes neuronals i *deep learning*

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LECTURE 1: Historical foundations of artificial neural networks

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1. Fundamentals of artificial neural networks

- 1 Historical notes.
- 2 Basic biological concepts.
- 3 The pioneering work of McCulloch & Pitts
- 4 Cognitive and computational implications. [MAT]
- 5 Lippmann networks.

[MAT] means that material is provided (in the *Racó*), but will not be seen in class or just superficially.

“The sciences do not try to explain, they hardly even try to interpret, they mainly make models. By a model is meant a mathematical construct which, with the addition of certain verbal interpretations, describes observed phenomena. The justification of such a mathematical construct is solely and precisely that it is expected to work - that is correctly to describe phenomena from a reasonably wide area. Furthermore, it must satisfy certain esthetic criteria - that is, in relation to how much it describes, it must be rather simple.”

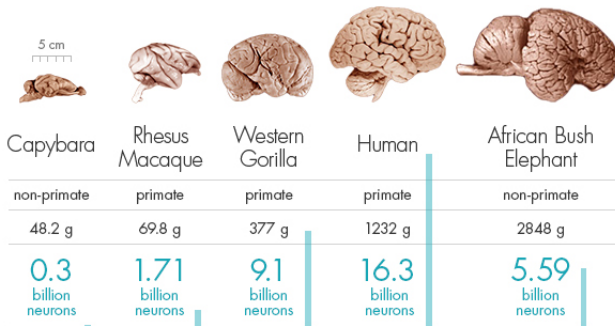
— John von Neumann.

NEURAL NETWORKS AND DEEP LEARNING

Basic biological concepts

BRAIN SIZE AND NEURON COUNT

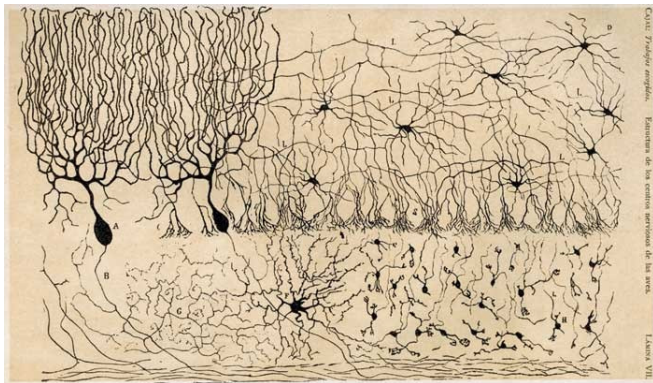
Cerebral cortex mass and neuron count for various mammals.



Comparison of the brains of various mammals.

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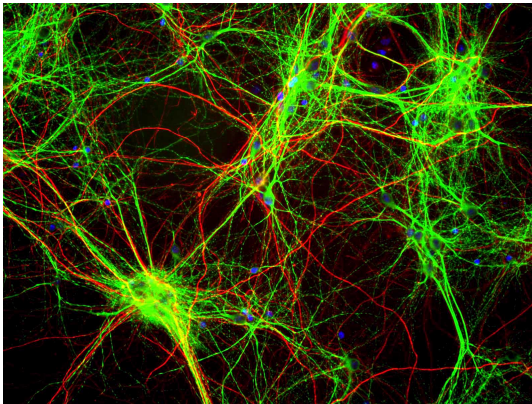
Basic biological concepts



Drawings made by Santiago Ramón y Cajal (awarded, together with the Italian Camillo Golgi, with the Nobel Prize for Medicine or Physiology in 1906); Purkinje, stellate and pyramidal neurons are seen.

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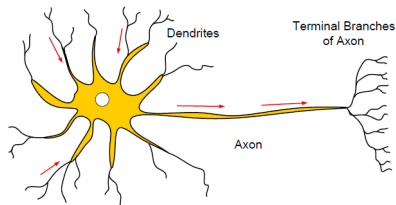
Basic biological concepts



Rat brain cell culture (standard tissue culture and immunofluorescence).

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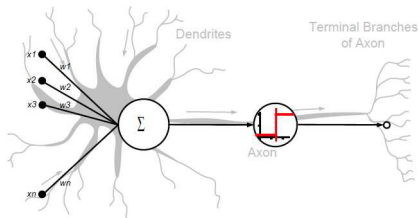
Basic biological concepts



- A **neuron** is an electrically excitable cell that processes and transmits information through electrical and chemical signals.
- A typical neuron is divided into three parts: the **soma** or cell body, **dendrites**, and **axon**.
- **Signals** from other neurons are received by the soma and dendrites; signals to other neurons are transmitted by the axon.
- A **synapse** is a contact between the axon of one neuron and a dendrite or soma of another. Synaptic signals may be excitatory or inhibitory.
- If the voltage received by a neuron over a period of time is large enough, it generates a brief **pulse** at the soma and propagates it along the axon.
- This pulse **activates** synaptic connections with other cells.

NEURAL NETWORKS AND DEEP LEARNING

Basic biological concepts



- the w 's correspond to the synaptic weights (excitatory or inhibitory);
- the multiplication represent the synapses processing of the inputs x 's;
- the summation is the combination of signals in the cell body (*net input*);
- the thresholding corresponds to the “decision” to fire (*activation*)

In the early days the term “artificial neural network” was motivated towards modeling networks of real neurons in the brain.

Timeline of Neural Networks

- 1943 McCulloch & Pitts's model
- 1949 Hebb's rule (further work by Caianiello)
- 1958 Perceptron (Rosenblatt)
- 1960 Adaline (Widrow & Hoff)
- 1969 *Perceptrons* (book), by Minsky & Papert
- 70s Self-Organizing Map models and Associative Memories (Kohonen'72, von der Malsburg'73, Amari'77, Grossberg'76, Anderson'72)
- 1980 Adaptive Resonance Theory models (Cohen & Grossberg)
- 1982 Energy-based models (Hopfield, Ackley, Hinton, Sejnowski)
- 1986 Backpropagation networks (Multilayer Perceptron) / *Parallel Distributed Processing* books (Rumelhart, Hinton, McClelland & Williams)
- 1988 Radial Basis Function networks (Broomhead & Lowe, Poggio & Girosi), Bidirectional associative memories (Kosko)
- 1995 Support Vector Machines (Boser, Cortes, Guyon & Vapnik)
- 1997 Recurrent networks (Jordan, Elman), Long short-term memory (Hochreiter & Schmidhuber)
- 1998 Convolutional networks (LeNet) by Le Cun
- 2006 Deep Belief Nets (Hinton, Osindero & Teh)
- 2009 Stacked autoencoder (many contributions)
- 2014 Generative Adversarial Networks (Goodfellow)

NEURAL NETWORKS AND DEEP LEARNING

The pioneering work of McCulloch & Pitts

A threshold-logic unit or TLU

$$y = g_{\text{HV}} \left(\sum_{j=1}^d w_j x_j - \theta \right), \quad g_{\text{HV}}(z) = \begin{cases} 1 & : z \geq 0 \\ 0 & : z < 0 \end{cases}$$

where $x_j \in \{0, 1\}$, $w_j \in \{-1, +1\}$, $\theta \in \mathbb{R}$.

Warren McCulloch (neurophysiologist and cybernetician) and Walter Pitts (logician and cognitive psychologist): “*A Logical Calculus of Ideas Immanent in Nervous Activity*” (1943).

NEURAL NETWORKS AND DEEP LEARNING

The pioneering work of McCulloch & Pitts

Notions of Boolean algebra

NOT		AND			OR			XOR		
x	F	x	y	F	x	y	F	x	y	F
0	1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	1	1	0	1	1
		1	0	0	1	0	1	1	0	1
		1	1	1	1	1	1	1	1	0

Boolean algebra is an algebraic structure on the set $0, 1$ that schematizes the logical operations.

NEURAL NETWORKS AND DEEP LEARNING

The pioneering work of McCulloch & Pitts

Proposition

Any logical function of n variables can be computed using a TLU neural network.

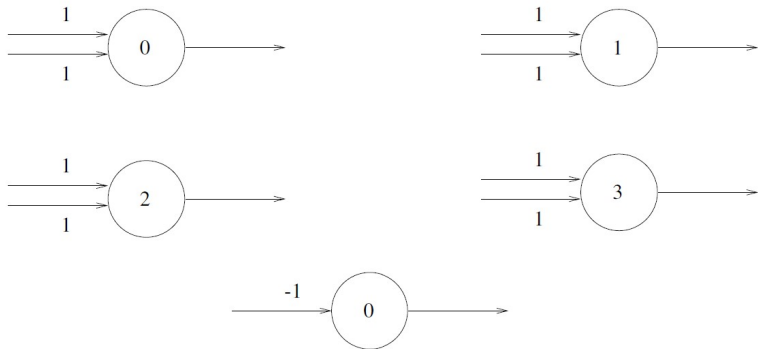
Proof:

- 1 Any logical function of n variables can be expressed only in terms of ANDs, ORs and NOTs (or NANDs, or NORs);
- 2 ANDs, ORs and NOTs (or NANDs, or NORs) are particular cases of a single TLU.

NEURAL NETWORKS AND DEEP LEARNING

The pioneering work of McCulloch & Pitts

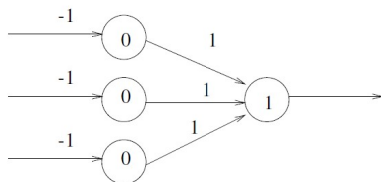
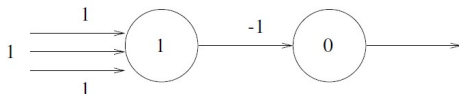
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NEURAL NETWORKS AND DEEP LEARNING

The pioneering work of McCulloch & Pitts

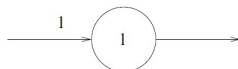
NANDs and NORs can also be implemented with TLUs:



NEURAL NETWORKS AND DEEP LEARNING

The pioneering work of McCulloch & Pitts

- Remark 1.** A digital computer is nothing more than a logical function with access to (finite) memory;
- Remark 2.** If we allow "feedback loops", it is possible to design an RS* bistable with one (or several) TLU neurons.
- Remark 3.** The last needed piece is a humble "delayer":

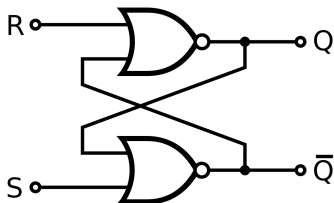


*also called a *flip/flop*: electronic circuit that can store one bit of memory; they are the basis of the SRAM (Static RAM) of modern hardware.

NEURAL NETWORKS AND DEEP LEARNING

The pioneering work of McCulloch & Pitts

S	R	Q_t	$\overline{Q}_t$
0	0	X	X
0	1	1	0
1	0	0	1
1	1	Q_{t-1}	$\overline{Q}_{t-1}$



NEURAL NETWORKS AND DEEP LEARNING

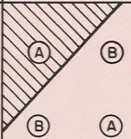
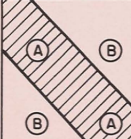
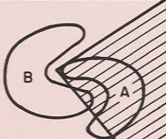
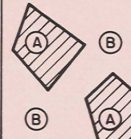
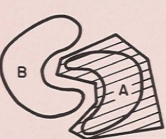
The pioneering work of McCulloch & Pitts

Observations

- 1 This was the first time that a model abstracted from the brain showed a real computing ability;
- 2 There is a heavy abstraction from biology, although inspired by works of Ramón y Cajal (on neural structures and of Sherrington (on synapses);
- 3 There is no learning and no distributed representation;
- 4 A Turing machine could in principle be simulated with such a network (and *vice versa*): these networks are therefore **Turing-equivalent**.

NEURAL NETWORKS AND DEEP LEARNING

Lippmann networks

Structure	Types of Decision Regions	Exclusive-or Problem	Classes with Meshed Regions	Region Shapes
One Layer 	Half-Plane			
Two Layers 	Typically Convex			
Three Layers 	Arbitrary			

NEURAL NETWORKS AND DEEP LEARNING

The Rosetta stone

Machine Learning	Statistics
model	model
parameter/weight	parameter/coefficient
train	fit
learn	infer/estimate
regression	regression
classification	discrimination/recognition
clustering	clustering
inputs/features/variables	independent or explanatory variables predictors, regressors
outputs/targets	dependent or response variables
instances/examples	individuals/observations
error/loss function	fit criterion/deviance
training/empirical error	resubstitution/in-sample error
test error	predictive/out-sample error
true/generalization error	risk

Careful with other words! transaction (means 'observation' in DBs), sample (means 'dataset' in MVS), attribute (means 'variable' in AI), ...

NEURAL NETWORKS AND DEEP LEARNING

Inductive bias

Complete the series! 2, 4, 6, 8, ...

Answer 1: 132 (model 1: $f(n) = n^4 - 10n^3 + 35n^2 - 48n + 24$)

Answer 2: 10 (model 2: $f(n) = 2n$)

How can we rule out the more complex one? (and many others)

- 1 Supply more “training” data: 2, 4, 6, 8, 10, 12, 14, ...
- 2 Regularize: add a penalty to higher-order terms
- 3 Reduce the hypothesis space (e.g. restrict to quadratic models)

NEURAL NETWORKS AND DEEP LEARNING

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NEURAL NETWORKS AND DEEP LEARNING

Why are all learning tasks stochastic?

We have a (complete) input data object $(\mathbf{x}, \mathbf{z})$ and a output data object t .

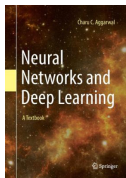
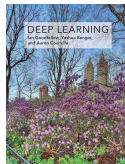
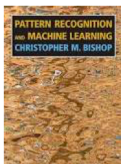
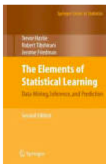
- 1 The true relation is $f^* : \mathcal{X} \times \mathcal{Z} \rightarrow \mathcal{T}$, that is $f^*(\mathbf{x}, \mathbf{z}) = t$.
- 2 When we measure data about f^* , in the form $D = \{(\mathbf{x}^n, t^n)\} \ n = 1, \dots, N$, we actually get the $\mathbf{x}$ portion of the input variables only.
- 3 Therefore, the relation between $\mathbf{x}$ and t becomes *stochastic*:

$\Rightarrow$ For every value of $\mathbf{x}^n$, there is a *distribution* of values for t^n .

NEURAL NETWORKS AND DEEP LEARNING

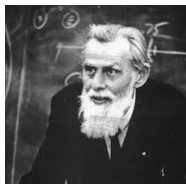
Recommended reading

- 1 *The Elements of Statistical Learning Data Mining, Inference, and Prediction*, Second Edition. Hastie, T.; Tibshirani, R.; Friedman, J.
- 2 *Pattern Recognition and Machine Learning*. Bishop, C.
- 3 *Pattern Classification*, 2nd Edition. Duda, R.O.; Hart, P.E.; Stork, D.G.
- 4 *Deep Learning*. Goodfellow, I.; Bengio, Y.; Courville, A.
- 5 *Deep Learning: Foundations and Concepts*. Bishop, C.
- 6 *Neural networks and deep learning: a textbook*. Aggarwal, C.C.

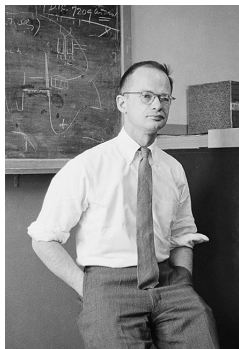


NEURAL NETWORKS AND DEEP LEARNING

Relevant characters for this lecture



Warren McCulloch
(1898-1969)



Walter Pitts
(1923-1969)



Santiago Ramón y Cajal
(1852-1934)